

Module 3 Lab 3 — Textbook Problem 2.2: Outliers in the Bridges Data

Course: Data Types, Storage, and Cleaning (Module 3) **Dataset:** `bridges.csv` — historical Pittsburgh-area bridges, one numeric attribute (`length`, in feet)

Task: Choose and *state* a plausibility assumption for bridge `length` **before** looking at which rows it flags, screen the column against that assumption, and then justify how the flagged rows were handled.

In [1]:

```
import pandas as pd
import numpy as np

df = pd.read_csv('bridges.csv')
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")
df.head()
```

Rows: 108, Columns: 13

	Id	river	location	erected	purpose	length	lanes	clear-g	t-or-d	material	span	rel-l	type
0	E1	M	3.0	1818	HIGHWAY	NaN	2.0	N	THROUGH	WOOD	SHORT	S	WOOD
1	E2	A	25.0	1819	HIGHWAY	1037.0	2.0	N	THROUGH	WOOD	SHORT	S	WOOD
2	E3	A	39.0	1829	AQUEDUCT	NaN	1.0	N	THROUGH	WOOD	NaN	S	WOOD
3	E5	A	29.0	1837	HIGHWAY	1000.0	2.0	N	THROUGH	WOOD	SHORT	S	WOOD
4	E6	M	23.0	1838	HIGHWAY	NaN	2.0	N	THROUGH	WOOD	NaN	S	WOOD

1. Look at the data before assuming anything

`length` is the only numeric attribute in this dataset (feet). Everything else (`river`, `purpose`, `material`, `span`, `type`, etc.) is categorical or an identifier. Before stating a plausibility rule, it's worth knowing (a) how much data is missing and (b) the overall shape of the distribution — a rule should be informed by what "normal" looks like for *these* bridges, not an arbitrary number.

In [2]:

```
missing = df['length'].isna().sum()
print(f"Missing 'length' values: {missing} of {len(df)} rows ({missing/len(df):.1%})")
print()
df['length'].describe()
```

Missing 'length' values: 27 of 108 rows (25.0%)

	length
count	81.000000
mean	1567.469136
std	747.491523
min	804.000000
25%	1000.000000
50%	1300.000000

75%	2000.000000
max	4558.000000

Two things worth noting:

- **27 rows (25%) have no recorded length at all.** These are *missing* values, not outliers — there is nothing implausible about a blank cell, so they are not part of the outlier screen below. They are a separate data-quality issue (handled at the end).
- Among the 81 rows that do have a length, the values range from 804 ft to 4,558 ft, with a median around 1,300 ft. No length is zero or negative, so there's no "obviously impossible" value the way Lab 2 had — this is exactly the subtler situation the assignment describes.

2. Stated assumption (before screening)

Context: these are 19th- and early-20th-century highway and railroad bridges in the Pittsburgh area, crossing the Allegheny, Monongahela, and Ohio rivers. `length` is the total structure length in feet.

Plausibility rule, stated before I look at what it flags:

A bridge `length` is plausible if it falls between **200 ft and 3,000 ft**.

- **Lower bound (200 ft):** a structure recorded in this dataset as a "bridge" spanning a river should be at least long enough to actually cross open water — a few dozen feet would suggest a culvert or a recording error, not a river bridge.
- **Upper bound (3,000 ft):** the vast majority of 19th-century wood/iron/steel truss and suspension bridges in this region were in the few-hundred-to-low-thousands-of-feet range. 3,000 ft is a generous ceiling — well above the 75th percentile (2,000 ft) — chosen so that only *unusually* long structures get flagged, not just the long tail of an already-long-tailed distribution.

Rows outside `[200, 3000]` are flagged below. I am committing to this rule **before** running the screen, per the assignment's instructions.

In [3]:

```

LOWER_BOUND = 200
UPPER_BOUND = 3000

has_length = df['length'].notna()
implausible = has_length & ((df['length'] < LOWER_BOUND) | (df['length'] > UPPER_BOUND))

print(f"Rows screened (non-missing length): {has_length.sum()}")
print(f"Rows flagged as implausible: {implausible.sum()}")
print()
flagged = df.loc[implausible, ['Id', 'river', 'erected', 'purpose', 'length', 'material', 'span',
                              'type']] \
    .sort_values('length', ascending=False)

flagged

```

```

Rows screened (non-missing length): 81
Rows flagged as implausible: 3

```

	Id	river	erected	purpose	length	material	span	type
31	E34	O	1888	RR	4558.0	STEEL	LONG	SIMPLE-T
44	E46	A	1897	RR	4000.0	STEEL	LONG	SIMPLE-T
104	E91	O	1975	HIGHWAY	3756.0	STEEL	LONG	ARCH

3. What got flagged

Three rows exceed the 3,000 ft upper bound: **E34** (4,558 ft), **E46** (4,000 ft), and **E91** (3,756 ft). No rows fall below the 200 ft lower bound — every recorded length is at least 804 ft, so the lower bound did not end up screening anything out for this particular dataset (it's still correct to state it, since a rule chosen *after* seeing that no row is short would be reasoning backwards).

In [4]:

```
# A closer look at *why* the three flagged rows are so long
flagged[['Id', 'river', 'erected', 'purpose', 'material', 'span', 'type', 'length']]
```

	Id	river	erected	purpose	material	span	type	length
31	E34	O	1888	RR	STEEL	LONG	SIMPLE-T	4558.0
44	E46	A	1897	RR	STEEL	LONG	SIMPLE-T	4000.0
104	E91	O	1975	HIGHWAY	STEEL	LONG	ARCH	3756.0

4. Handling the flagged rows, and why

I am **not deleting or capping** these three rows. Here's the reasoning:

- All three (E34, E46, E91) have `span == LONG` and `material == STEEL`. That combination is exactly what you'd expect to *legitimately* produce a length far outside the typical wood/iron truss range: a long-span steel bridge (e.g. a railroad bridge crossing the Ohio River, or the 1975 highway bridge E91) is structurally capable of — and expected to — run several thousand feet, especially for a multi-span railroad crossing.
- In other words, my *statistical* rule (3,000 ft ceiling) correctly flagged these as unusual relative to the rest of the dataset, but the *domain* evidence in the other columns (`span`, `material`, `purpose`) supports treating them as real, valid measurements rather than data-entry errors — there's no column where a "4,558" could plausibly be a typo for something in-range (e.g. a decimal-point slip would need to be off by 10x, which isn't a typical typo pattern).
- **Decision:** keep all three rows unchanged in the analysis dataset, but retain the boolean `length_flag` column below so any downstream analysis (e.g. an average-length-by-material calculation) can choose to exclude or separately weight them if their scale would distort a mean or a chart.

Separately, the **27 missing length values** are handled differently, since they are a missing-data problem rather than an outlier problem: I leave them as `NaN` (not imputed to 0 or to the mean) so that any later aggregation on `length` correctly excludes them via `.dropna()` / `.mean()` rather than being silently biased by a fabricated fill value.

In [5]:

```
df['length_flag'] = implausible
print("Value counts for the new 'length_flag' column:")
print(df['length_flag'].value_counts())
print()
kept = (~df['length_flag']).sum()
print(f"Rows kept as-is (not flagged, including the 27 with missing length): {kept} / {len(df)}")
```

```
Value counts for the new 'length_flag' column:
length_flag
False      105
True         3
Name: count, dtype: int64
```

```
Rows kept as-is (not flagged, including the 27 with missing length): 105 / 108
```

5. Summary

- **Assumption (stated first):** a plausible bridge length is between 200 ft and 3,000 ft, based on the physical need to span a river and the typical scale of 19th-century truss/suspension bridges in this region.
- **Screen:** applied to the 81 non-missing `length` values; flagged 3 of them (E34, E46, E91), all > 3,000 ft.
- **Handling:** kept all three flagged rows — the domain columns (`span=LONG` , `material=STEEL`) support them as genuine long-span steel bridges rather than data-entry errors — and added a `length_flag` column so they can be excluded from length-sensitive summary statistics later if needed. The 27 missing lengths are left as `NaN` , since they are a separate missing-data issue, not an outlier.

GitHub repo: `TODO` — paste your `module3/` GitHub repo URL here before submitting